

Infinite Quantum Group Symmetry of Fields in Massive 2D Quantum Field Theory

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Abstract

Starting from a given S-matrix of an integrable quantum field theory in $1+1$ dimensions, and knowledge of its on-shell quantum group symmetries, we describe how to extend the symmetry to the space of fields. This is accomplished by introducing an adjoint action of the symmetry generators on fields, and specifying the form factors of descendents. The braiding relations of quantum field multiplets is shown to be given by the universal R -matrix. We develop in some detail the case of infinite dimensional Yangian symmetry. We show that the quantum double of the Yangian is a Hopf algebra deformation of a level zero Kac-Moody algebra that preserves its finite dimensional Lie subalgebra. The fields form infinite dimensional Verma-module representations; in particular the energy-momentum tensor and isotopic current are in the same multiplet.

1 Introduction

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Smirnov, in Introduction to Quantum Groups and Integrable Massive Models of Quantum Field Theory, Nankai Lectures in Mathematical Physics, Mo-Lin Ge and Bao-Heng Zhao (eds.), World Scientific, Singapore (1990) A. N. Kirillov and F. A. Smirnov, Phys. Lett. 188B (1987) 506; Zap. Nauch. Semin. LOMI 164 (1987) 80.

2 Method

L. D. Faddeev, N. Yu. Reshetikhin, and L. A. Takhtajan, Quantization of Lie Groups and Lie Algebras, 1989 preprint. P. Mathieu, Extended Classical Conformal Algebras and the Second Hamiltonian Structure of Lax Equations, Universit'e Laval Preprint, 1988 V. Fateev and S. Lykhanov, Landau Institute preprint (in Russian).

$$L(\theta) = \sum_{i=1}^n \ell(x_i, \theta) \tag{1}$$

Equation 1 is included to keep the sample paper-like while remaining small.

3 Conclusion

G. E. Andrews, q-series: Their Development and Application in Analysis, Number Theory, Combinatorics, Physics, and Computer Algebra, AMS Regional Conference Series 66 (1986). M. B. Green, J. H. Schwarz, and E. Witten, Superstring Theory I, Cambridge University Press (1987).

L. J. Romans, A New Family of Dual Models ('q-strings') , Univ. Southern Cal. Preprint USC-88/HEP014. I. B. Frenkel and N. Jing, Vertex Representations of Quantum Affine Algebras , Yale Math. Preprint, (1988)